Geometry and arithmetic of torsors under tori over abelian varieties

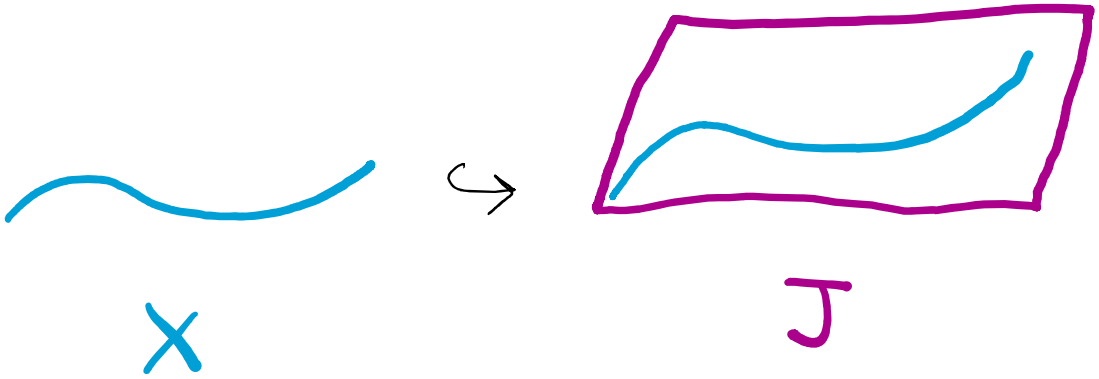
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Introduction

Abel–Jacobi embedding

$$X(\mathbb{Q}) = X \cap J(\mathbb{Q})$$

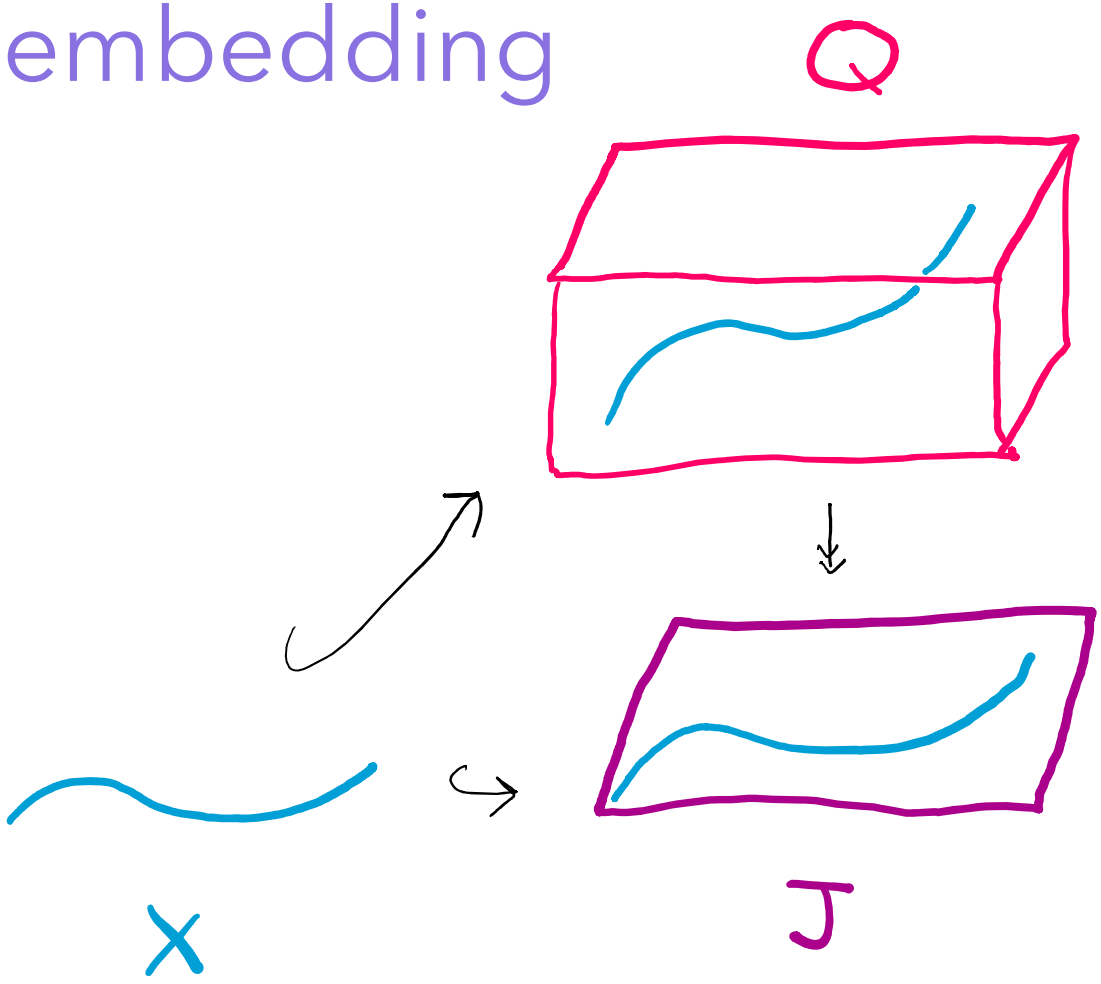
- Vojta's proof of Mordell.
- Chabauty's method.
- Manin–Mumford Conjecture.
- Mordell–Lang Conjecture.



Quadratic Albanese embedding

$$X(\mathbb{Q}) = X \cap Q(\mathbb{Q})$$

- Quadratic Chabauty.
- Manin–Mumford Conjecture.
- Hyperbolic curves.
- …



AT varieties

In which we meet the principal players in our tale.

AT varieties

Let k be a perfect field. An AT variety is a variety P/k such that $P_{\bar{k}}$ is isomorphic to a $\mathbb{G}_{\mathrm{m}}^{r}$-torsor over an abelian variety.

Example 1: any semiabelian variety is an AT variety.

Example 2: if E/k is an elliptic curve, then the $\mathbb{G}_{\mathrm{m}}$-torsor associated to $\mathcal{O}(\infty)$ is an AT variety.

Claim: P in Example 2 is not a semiabelian variety.

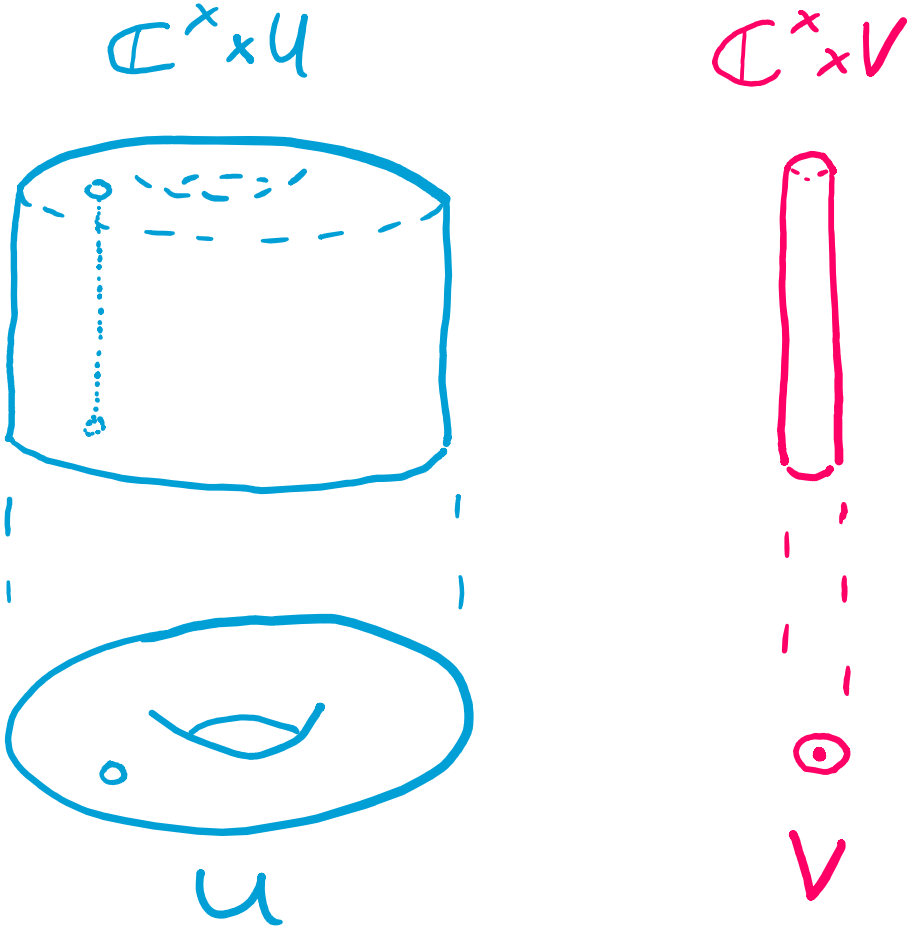
P is not semiabelian

Work analytically over $\mathbb{C}$.

Let $U = E \setminus \{\infty\}$ and V be a small neighbourhood of ∞.

$P = (\mathbb{C}^\times \times U) \cup (\mathbb{C}^\times \times V)$, with gluing data on $\mathbb{C}^\times \times (U \cap V)$ given by

$$(w, z) \mapsto (t(z) \cdot w, z)$$

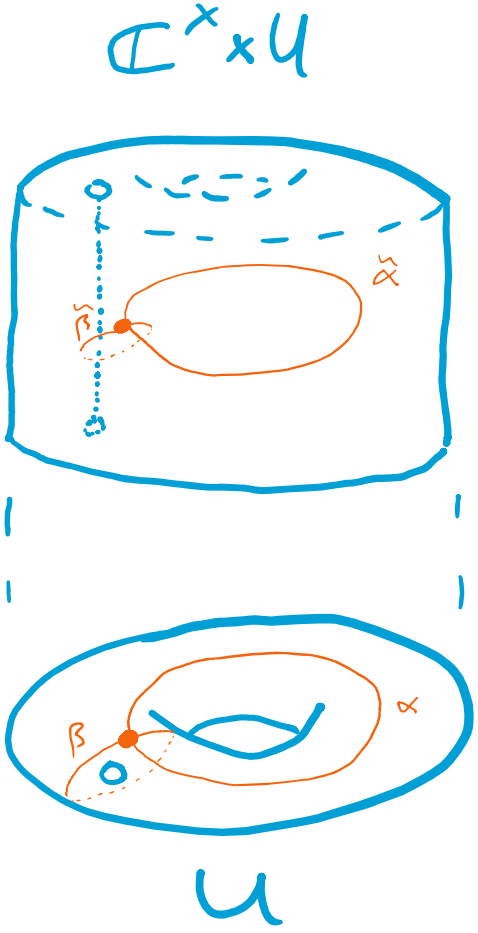


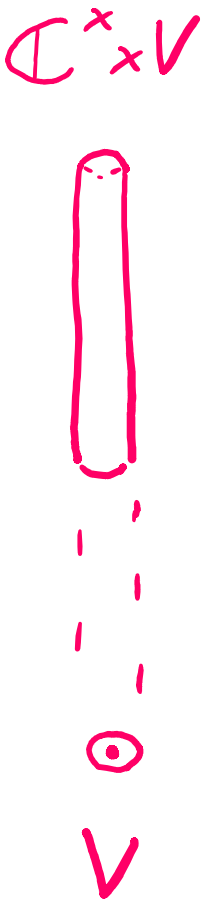
P is not semiabelian

Let α and β be the standard generators of $\pi_1(U)$.

Let $\tilde{\alpha} = (1, \alpha)$ and $\tilde{\beta} = (1, \beta)$ be the obvious lifts to $\mathbb{C}^\times \times U$.

The commutator $[\alpha, \beta]$ is a small loop around ∞, and $[\tilde{\alpha}, \tilde{\beta}]$ is its lift to $\mathbb{C}^\times \times U$.





P is not semiabelian

In local coordinates on $\mathbb{C}^\times \times U$, $[\tilde{\alpha}, \tilde{\beta}]$ is given by

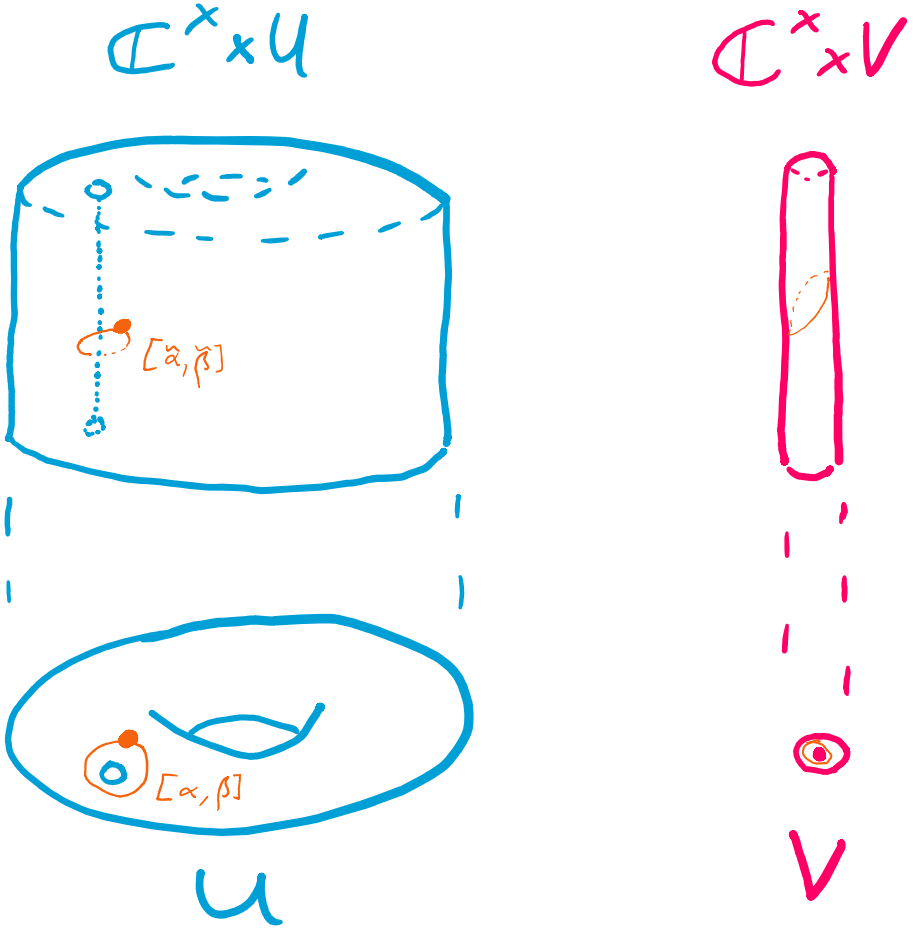
$$\tau \mapsto (1, \epsilon \exp(2\pi i \tau))$$

In local coordinates on $\mathbb{C}^\times \times V$, $[\tilde{\alpha}, \tilde{\beta}]$ is given by

$$\tau \mapsto (\epsilon \exp(2\pi i \tau), \epsilon \exp(2\pi i \tau))$$

So $[\tilde{\alpha}, \tilde{\beta}]$ is non-trivial in $\pi_1(P)$.

$\therefore \pi_1(P)$ is non-abelian!



Fundamental groups of AT varieties

If P is a pointed AT variety over $\mathbb{C}$, then:

- $\pi_1(P)$ is 2-step nilpotent.
- If P is a $\mathbb{G}_{\mathrm{m}}^r$-torsor over an abelian variety A, then $\pi_1(P)$ is a central extension of $\mathrm{H}_1(A)$ by $\mathbb{Z}(1)^r$.
- If P is the $\mathbb{G}_{\mathrm{m}}$-torsor corresponding to a line bundle L on A, then the commutator pairing $\bigwedge^2 \mathrm{H}_1(A) \to \mathbb{Z}(1)$ is $2\pi i \cdot c(L)$.
- P is semiabelian $\Leftrightarrow$ $\pi_1(P)$ is abelian.

Cubical structures

In which we discover a hidden algebraic structure on AT varieties.

Cubical structures: definition

If P and Q are G-torsors over B, then $P \otimes Q$ is the torsor obtained from $P \times_B Q$ by pushing out along $+: G \times G \to G$.

If A is an abelian group and $I \subseteq \{1,2,3\}$, let $m_I: A^3 \to A$ be the map $(x_1, x_2, x_3) \mapsto \sum_{i \in I} x_i$.

If P is a G-torsor over A, then define its theta-torsor to be

$$\Theta_3(P) := \bigotimes_{I \subseteq \{1,2,3\}} m_I^* P^{\otimes (-1)^{3-\#I}}$$

Cubical structures: definition

Definition (Breen):

A cubical structure on P is a trivialisation of $\Theta_3(P)$ satisfying certain conditions (associativity, unitality and commutativity).

Theorem (Breen):

Every pointed AT variety possesses a unique cubical structure. These cubical structures are functorial with respect to morphisms of AT varieties.

The cubical structure comes from the theorem of the cube!

Relation to Néron–Tate heights

Let $P/\mathbb{Q}_p$ be a pointed $\mathbb{G}_{\mathbf{m}}$-torsor over an abelian variety A.

Breen proves that P has a Néron model $\mathcal{P}$ over $\mathbb{Z}_p$. $\mathcal{P}$ is a Zariski-locally trivial cubical $\mathcal{G}_{\mathbf{m}}$-torsor over $\mathcal{A}$, where $\mathcal{G}_{\mathbf{m}}$ and $\mathcal{A}$ are the Néron models of $\mathbb{G}_{\mathbf{m}}$ and A.

$\mathcal{P}$ is not finite type.

Relation to Néron–Tate heights

If $i\colon \mathrm{Spec}(\mathbb{F}_p) \hookrightarrow \mathrm{Spec}(\mathbb{Z}_p)$ is the inclusion of the special point, then there is a homomorphism of sheaves $\mathcal{G}_\mathrm{m} \to i_*\mathbb{Z}$.

Theorem (Breen):

1. There is a unique morphism $\lambda\colon \mathcal{P} \to i_*\mathbb{Q}$ compatible with the cubical structures via the homomorphism $\mathcal{G}_\mathrm{m} \to i_*\mathbb{Q}$.
2. If L is the rigidified line bundle corresponding to P, and if $||\cdot||\colon L(\mathbb{Q}_p) \to \mathbb{R}$ is its canonical p-adic metric, then
$$\lambda = v_p||\cdot||$$

Unlikely intersections

In which we resume our study of the geometry of complex AT varieties.

Unlikely intersections: setup

Complex AT varieties P are a natural setting for unlikely intersections problems.

- Weakly special subvarieties = AT subvarieties.
- Special points: fix a base point on $P \Rightarrow$ define a torsion subset $P_{\mathrm{tors}} \subset P$.

Manin–Mumford problem: if $X \subseteq P$ is closed, are all points in $X \cap P_{\mathrm{tors}}$ accounted for by finitely many AT subvarieties of X?

Torsion subsets

Assume for simplicity that P is a *symmetric* $\mathbb{G}_{\mathrm{m}}^r$-torsor over an abelian variety A, and the base point $\tilde{0} \in P(\mathbb{C})$ lies over $0 \in A(\mathbb{C})$.

Then $[n]^* P \cong P^{\otimes n^2}$, so there is an endomorphism

$$\beta_n \colon P \to P$$

lying over $[n] \colon A \to A$ and satisfying $\beta_n(\lambda \tilde{x}) = \lambda^{n^2} \beta_n(\tilde{x})$.

The torsion subset is $P_{\mathrm{tors}} \coloneqq \bigcup_n \ker(\beta_n)$.

μ_∞^r-torsor over A_{tors}.

Manin–Mumford for AT varieties

Theorem (B., "Manin–Mumford for curves in AT varieties"):

Let P be a complex pointed AT variety, and let $X \subseteq P$ be a closed irreducible subvariety of dimension 1. If X is not an AT subvariety of P, then $X \cap P_{\mathrm{tors}}$ is finite.

Manin–Mumford for AT varieties

Proof 1:

- If P is not a $\mathbb{G}_{\mathrm{m}}$-torsor over an elliptic curve, then P admits a morphism to a semiabelian variety B of dimension ≥ 2.
  $\Rightarrow$ reduce to Manin–Mumford for semiabelian varieties.
- If P is a $\mathbb{G}_{\mathrm{m}}$-torsor over an elliptic curve E, then we can produce a family of semiabelian surfaces over E.
  $\Rightarrow$ reduce to relative Manin–Mumford for semiabelian surfaces (Bertrand–Masser–Pillay–Zannier).

Manin–Mumford for AT varieties

Proof 2 (unpublished):

Use the Pila–Zannier strategy.

- Hain–Zucker: $P \cong \pi_1(P)\backslash\pi_1(P)_{\mathbb{C}}/\mathrm{F}^0$ uniformised by $\pi_1(P)_{\mathbb{C}}/\mathrm{F}^0$.
- P_{tors} is the image of $\pi_1(P)_{\mathbb{Q}}$.
- The universal covering $\pi_1(P)_{\mathbb{C}}/\mathrm{F}^0 \to P$ is definable when restricted to a definable fundamental domain.
- Functional transcendence (Ax–Lindemann): if $W \subseteq \pi_1(P)_{\mathbb{C}}/\mathrm{F}^0$ is closed and irred., then $\pi(W)^{\mathrm{Zar}} \subseteq P$ is an AT subvariety.
- Masser-style lower bound on Galois orbits in P_{tors}.

Rational points on curves

In which we connect these new varieties to Diophantine questions.

Quadratic Albanese variety

Theorem (B.):

Let X be a smooth curve over a perfect field k. Then there exists an AT variety Q and a morphism

$$q: X \to Q$$

which is initial among all morphisms from X into AT varieties.

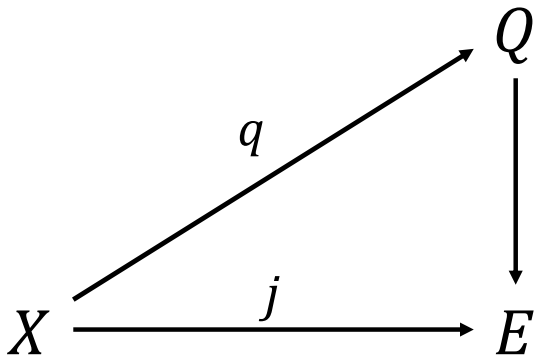
Q is called the quadratic Albanese variety of X.

q is called the quadratic Albanese map for X.

Quadratic Albanese variety: example

If $X = E \setminus \{\infty\}$, then the quadratic Albanese variety of X is the $\mathbb{G}_{\mathrm{m}}$-torsor Q over E corresponding to $\mathcal{O}(\infty)$.

j^*Q is the $\mathbb{G}_{\mathrm{m}}$-torsor corresponding to $\mathcal{O}_X$, i.e. is the trivial torsor. So j lifts to a morphism $q\colon X \to Q$. This lift is the quadratic Albanese map.



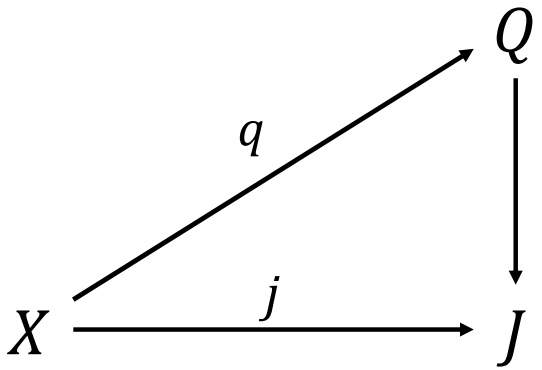
Quadratic Albanese variety: example

If X is smooth and projective with $X(k) \neq \emptyset$, then

$$\ker\left(\mathrm{Pic}(J_{\bar{k}}) \to \mathrm{Pic}(X_{\bar{k}})\right) = \ker(\mathrm{NS}(J_{\bar{k}}) \to \mathrm{NS}(X_{\bar{k}}))$$

is finitely generated free.

The quadratic Albanese variety of X is the fibre product of the $\mathbb{G}_{\mathrm{m}}$-torsors corresponding to a basis.



Rational points on curves

For $X/\mathbb{Q}$ a nice curve, embedded in its quadratic Albanese variety Q, we have

$$X(\mathbb{Q}) = X \cap Q(\mathbb{Q})$$

Assuming for simplicity that $X(\mathbb{Q}) \neq \emptyset$ and $\mathrm{NS}\left(J_{\overline{\mathbb{Q}}}\right) = \mathrm{NS}(J)$, then $Q(\mathbb{Q})$ is a $(\mathbb{Q}^\times)^r$-torsor over $J(\mathbb{Q})$.

Rational points on curves (better)

Better: consider integral models

- $\mathcal{X}$, the minimal regular model of X.
- $\mathcal{J}$, the Néron model of J.
- $\mathcal{G}_{\mathrm{m}}$, the Néron model of $\mathbb{G}_{\mathrm{m}}$.
- $\mathcal{Q}$, the Néron model of Q. It is a $\mathcal{G}_{\mathrm{m}}^{r}$-torsor over $\mathcal{J}$.
- $X \hookrightarrow Q$ spreads out to $\mathcal{X}_{\mathrm{sm}} \to \mathcal{Q}$; image is contained in a finite union of components $\mathcal{Q}_0 \subset \mathcal{Q}$.

$$\text{Then } X(\mathbb{Q}) = \mathcal{X}_{\mathrm{sm}}(\mathbb{Z}) = X \cap \underbrace{\mathcal{Q}_0(\mathbb{Z})}.$$

Finite union of $(\mathbb{Z}^{\times})^{r}$-torsors over cosets of $\mathcal{J}^{0}(\mathbb{Z})$.

Geometric quadratic Chabauty

If p is a prime, then we have

$$X(\mathbb{Q}) \subseteq X(\mathbb{Q}_p) \cap \overline{Q_0(\mathbb{Z})}$$

where the intersection and closure are taken inside $Q(\mathbb{Q}_p)$.

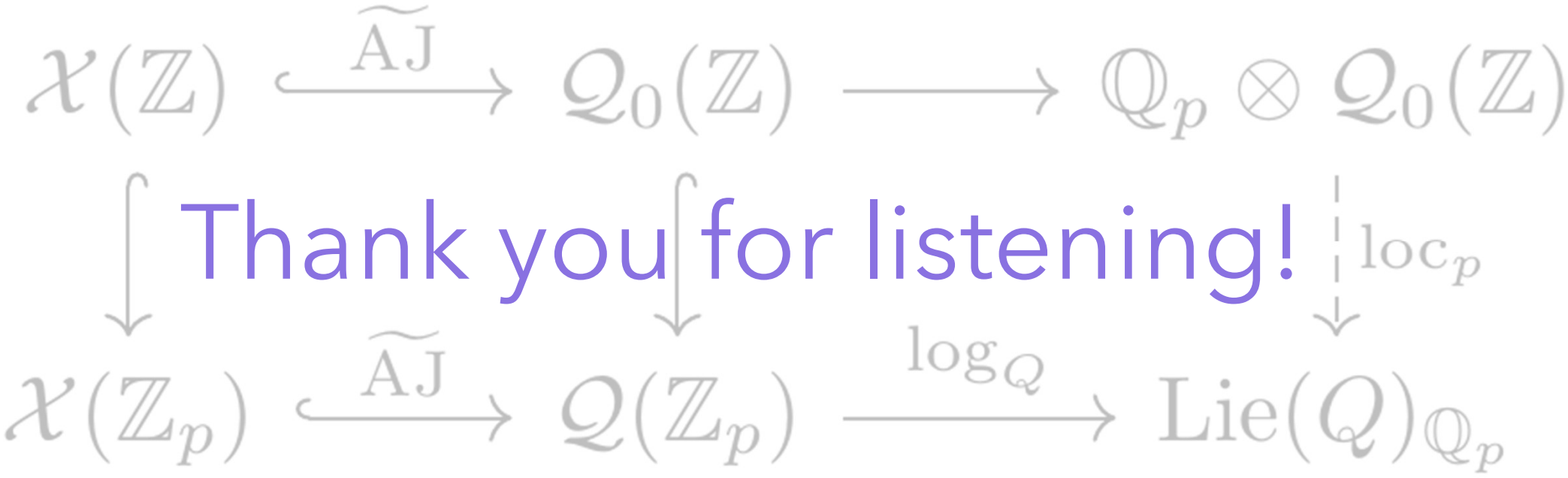
- The right-hand side is essentially the locus studied in geometric quadratic Chabauty (Edixhoven–Lido).
- It is finite if $\mathrm{rk}(J(\mathbb{Q})) < \dim(Q)$ (Edixhoven–Lido).
- Can modify this description to similarly describe the cohomological quadratic Chabauty locus (B., in progress).

Stoll's Conjecture

Theorem (B.):

If X is a nice curve over $\mathbb{Q}$ whose Jacobian has rank 0, then there is a finite subscheme $Z \subset X$ such that for each p of good reduction, $Z(\mathbb{Q}_p) \subset X(\mathbb{Q}_p)$ is the quadratic Chabauty locus.

- Also applies to hyperbolic curves.
- Stoll conjectured that if $\mathrm{rk}(J(\mathbb{Q})) \leq g - 2$, then there should be a finite Z such that $Z(\mathbb{Q}_p)$ contains the Chabauty locus.



Thank you for listening!

Unexpected points on once-punctured elliptic curves

joint work with Jennifer Balakrishnan

Once-punctured elliptic curves

$X = E \setminus \{\infty\}$ with E an elliptic curve of rank 0.

$\Rightarrow$ exists a finite subscheme $Z \subset X$ with

$$X(\mathbb{Z}_p)_{\mathrm{QC}} = Z(\mathbb{Q}_p)$$

for all primes p of good reduction.

Theorem (Bianchi, '20):

Any point in $X(\mathbb{Z}_p)_{\mathrm{QC}}$ is torsion.

Theorem (Balakrishnan–B., in progress):

The order of any point in $X(\mathbb{Z}_p)_{\mathrm{QC}}$ is in $\{2, 3, \ldots, 10, 12, 15\}$ or > 150.